Mid-term Exam II

Review of Modules 4–6

FINE 3010, Financial Management
Prof. Renping Li, Tulane University

Mid-term Exam II

- Monday 10/26, in class, 50 minutes, 100 points, covers **Modules 4–6** only
- Questions 1–10 multiple choice (6 points each), Questions 11–12 free response (20 points each)
- Show your work on free response, partial credit for correct steps
- Formula sheet provided, plus your **financial calculator** and one self-prepared cheat sheet (one page, double-sided, A4)

Set the Number of Digits to at Least 4!

- On the TI BA II Plus, press
  - **2nd**
  - .
  - 4
  - **ENTER**

Set the Number of Compounding Periods per Year to 1!

- On the TI BA II Plus, press
  - **2nd**
  - **I/Y** (**P/Y**)
  - 1
  - **ENTER**

Set the Number of Compounding Periods per Year to 1!

- On the TI BA II Plus, press
  - **2nd**
  - **I/Y** (**P/Y**)
  - 1
  - **ENTER**

You must do this!

Financial Calculators, Use Case One



- Financial calculators solve any one of **N**, **I/Y**, **PV**, **PMT**, and **FV** from the other four

Financial Calculators, Use Case One



- Financial calculators solve any one of **N**, **I/Y**, **PV**, **PMT**, and **FV** from the other four
- Your financial calculator is programmed to solve this equation

$$\text{PV} + \left[\frac{\text{PMT}}{(1+I)^1} + \frac{\text{PMT}}{(1+I)^2} \cdots + \frac{\text{PMT}}{(1+I)^N}\right] + \frac{\text{FV}}{(1+I)^N} = 0$$

Financial Calculators, Use Case Two

- Financial calculators also compute the NPV and the IRR of any stream of cash flows $CF_0, CF_1, \ldots, CF_N$ on the cash flow worksheet

Financial Calculators, Use Case Two

- Financial calculators also compute the NPV and the IRR of any stream of cash flows $\mathrm{CF}_0, \mathrm{CF}_1, \ldots, \mathrm{CF}_N$ on the cash flow worksheet
- **NPV**, your financial calculator computes this value at the I you enter

$$\mathrm{NPV} = \mathrm{CF}_0 + \frac{\mathrm{CF}_1}{(1+I)^1} + \frac{\mathrm{CF}_2}{(1+I)^2} + \cdots + \frac{\mathrm{CF}_N}{(1+I)^N}$$

Financial Calculators, Use Case Two

- Financial calculators also compute the NPV and the IRR of any stream of cash flows $\mathrm{CF}_0, \mathrm{CF}_1, \ldots, \mathrm{CF}_N$ on the cash flow worksheet
- **NPV**, your financial calculator computes this value at the I you enter

$$\mathrm{NPV} = \mathrm{CF}_0 + \frac{\mathrm{CF}_1}{(1+I)^1} + \frac{\mathrm{CF}_2}{(1+I)^2} + \cdots + \frac{\mathrm{CF}_N}{(1+I)^N}$$

- **IRR**, your financial calculator solves this equation for I

$$\mathrm{CF}_0 + \frac{\mathrm{CF}_1}{(1+I)^1} + \frac{\mathrm{CF}_2}{(1+I)^2} + \cdots + \frac{\mathrm{CF}_N}{(1+I)^N} = 0$$

Use Case Two, Keys for NPV and IRR

NPV

Enter	Keys
	CF, **2nd**, **CLR Work**
CF_0	**ENTER**, ↓
CF_1	**ENTER**, ↓, ↓
⋮	⋮
CF_N	**ENTER**
	NPV (the calculator asks for I)
I	**ENTER**, ↓
	CPT
NPV	

IRR

Enter	Keys
	CF, **2nd**, **CLR Work**
CF_0	**ENTER**, ↓
CF_1	**ENTER**, ↓, ↓
⋮	⋮
CF_N	**ENTER**
	IRR (shows 0)
	CPT
IRR	

Module 4, Future Value and Present Value

- A dollar today is worth more than a dollar tomorrow

Module 4, Future Value and Present Value

- A dollar today is worth more than a dollar tomorrow
- **Future value**, the amount to which an investment will grow after earning interest, $\text{FV} = \text{PV} \times (1 + r)^T$

Module 4, Future Value and Present Value

- A dollar today is worth more than a dollar tomorrow
- **Future value**, the amount to which an investment will grow after earning interest, $\text{FV} = \text{PV} \times (1 + r)^T$
- **Compound interest**, interest on interest, the interest earned is $\text{PV} \times (1 + r)^T - \text{PV}$

Module 4, Future Value and Present Value

- A dollar today is worth more than a dollar tomorrow
- **Future value**, the amount to which an investment will grow after earning interest, $\text{FV} = \text{PV} \times (1 + r)^T$
- **Compound interest**, interest on interest, the interest earned is $\text{PV} \times (1 + r)^T - \text{PV}$
- **Present value**, the value today of a future cash flow, $\text{PV} = \text{FV}/(1 + r)^T$

Example, Future Value

- Suppose you earn an annual return of 6%. How many years will it take until your money doubles?

Example, Future Value

- Suppose you earn an annual return of 6%. How many years will it take until your money doubles?
- That is, what is the T such that $(1 + 0.06)^T = 2$?

Example, Future Value

- Suppose you earn an annual return of 6%. How many years will it take until your money doubles?
- That is, what is the T such that $(1 + 0.06)^T = 2$?



$$N = ?, \quad I = 6\%$$

Example, Future Value

- Suppose you earn an annual return of 6%. How many years will it take until your money doubles?
- That is, what is the T such that $(1 + 0.06)^T = 2$?



$$N = ?, \quad I = 6\%$$

- -1 **PV**, 6 **I/Y**, 0 **PMT**, 2 **FV**, then **CPT** **N**

Example, Future Value

- Suppose you earn an annual return of 6%. How many years will it take until your money doubles?
- That is, what is the T such that $(1 + 0.06)^T = 2$?



$$N = ?, \quad I = 6\%$$

- -1 **PV**, 6 **I/Y**, 0 **PMT**, 2 **FV**, then **CPT** **N**
- The output is 11.9, about 11.9 years, because $(1 + 0.06)^{11.9} \approx 2$

Recap, Three Kinds of Returns

Return	Holding period return	Annualized return
Percentage growth of an asset's value over some period of time	Percentage growth over the whole period an asset is owned, longer or shorter than a year	Percentage growth per year, a geometric mean over the T years of the holding period

Module 4, Holding Period and Annualized Returns

- $1 + r_{\text{holding period}} = (1 + r_1) \times (1 + r_2) \times \cdots \times (1 + r_T)$

Module 4, Holding Period and Annualized Returns

- $1 + r_{\text{holding period}} = (1 + r_1) \times (1 + r_2) \times \cdots \times (1 + r_T)$
- $(1 + r_{\text{annualized}})^T = 1 + r_{\text{holding period}} \quad \rightarrow \quad r_{\text{annualized}} = (1 + r_{\text{holding period}})^{1/T} - 1$

Module 4, Holding Period and Annualized Returns

- $1 + r_{\text{holding period}} = (1 + r_1) \times (1 + r_2) \times \cdots \times (1 + r_T)$
- $(1 + r_{\text{annualized}})^T = 1 + r_{\text{holding period}} \quad \rightarrow \quad r_{\text{annualized}} = (1 + r_{\text{holding period}})^{1/T} - 1$

- E.g., a stock returns 80% in the first year and -20% in the second year
  - Holding period return, $1.80 \times 0.80 - 1 = 44\%$

Module 4, Holding Period and Annualized Returns

- $1 + r_{\text{holding period}} = (1 + r_1) \times (1 + r_2) \times \cdots \times (1 + r_T)$
- $(1 + r_{\text{annualized}})^T = 1 + r_{\text{holding period}} \quad \rightarrow \quad r_{\text{annualized}} = (1 + r_{\text{holding period}})^{1/T} - 1$

- E.g., a stock returns 80% in the first year and -20% in the second year
  - Holding period return, $1.80 \times 0.80 - 1 = 44\%$
  - Annualized return, $1.44^{1/2} - 1 = 20\%$, *not* $(80\% - 20\%)/2 = 30\%$

Module 4, Perpetuities and Annuities

- **Perpetuity**, a constant stream of cash flows that never ends, $\text{PV} = C/r$

Module 4, Perpetuities and Annuities

- **Perpetuity**, a constant stream of cash flows that never ends, $\text{PV} = C/r$
- **Growing perpetuity**, grows at g forever, $\text{PV} = C/(r-g)$

Module 4, Perpetuities and Annuities

- **Perpetuity**, a constant stream of cash flows that never ends, $\text{PV} = C/r$
- **Growing perpetuity**, grows at g forever, $\text{PV} = C/(r-g)$
- **Annuity**, a constant stream of cash flows that ends after N periods

$$\text{PV of annuity} = C \times \frac{1}{r}\left(1 - \frac{1}{(1+r)^N}\right)$$

Module 4, Perpetuities and Annuities

- **Perpetuity**, a constant stream of cash flows that never ends, $\text{PV} = C/r$
- **Growing perpetuity**, grows at g forever, $\text{PV} = C/(r-g)$
- **Annuity**, a constant stream of cash flows that ends after N periods

$$\text{PV of annuity} = C \times \frac{1}{r}\left(1 - \frac{1}{(1+r)^N}\right)$$

- On the calculator, N **N**, r **I/Y**, C **PMT**, 0 **FV**, then **CPT** **PV**

Module 4, Perpetuities and Annuities

- **Perpetuity**, a constant stream of cash flows that never ends, PV $= C/r$
- **Growing perpetuity**, grows at g forever, PV $= C/(r-g)$
- **Annuity**, a constant stream of cash flows that ends after N periods

$$\text{PV of annuity} = C \times \frac{1}{r}\left(1 - \frac{1}{(1+r)^N}\right)$$

- On the calculator, N **N**, r **I/Y**, C **PMT**, 0 **FV**, then **CPT** **PV**
- Timing, the first payment is at $t = 1$. Due (first payment at $t = 0$), multiply by $(1 + r)$. Delayed (first payment at $t > 1$), multiply by $1/(1 + r)^{t-1}$

Module 4, Inflation

- **Inflation**, the rate at which prices as a whole are increasing (the growth rate of the CPI)

Module 4, Inflation

- **Inflation**, the rate at which prices as a whole are increasing (the growth rate of the CPI)
- **Nominal return**, the rate at which money invested grows. **Real return**, the rate at which purchasing power increases

Module 4, Inflation

- **Inflation**, the rate at which prices as a whole are increasing (the growth rate of the CPI)
- **Nominal return**, the rate at which money invested grows. **Real return**, the rate at which purchasing power increases

$$(1 + \text{Real return}) = \frac{(1 + \text{Nominal return})}{(1 + \text{Inflation rate})}$$

$$\text{Approximation,} \quad \text{Real return} \approx \text{Nominal return} - \text{Inflation rate}$$

Module 4, Inflation

- **Inflation**, the rate at which prices as a whole are increasing (the growth rate of the CPI)
- **Nominal return**, the rate at which money invested grows. **Real return**, the rate at which purchasing power increases

$$(1 + \text{Real return}) = \frac{(1 + \text{Nominal return})}{(1 + \text{Inflation rate})}$$

$$\text{Approximation,} \quad \text{Real return} \approx \text{Nominal return} - \text{Inflation rate}$$

- E.g., a nominal return of 6% and an inflation rate of 2%, real return $= 1.06/1.02 - 1 = 3.9\%$, approximately $6\% - 2\% = 4\%$

Practice Problem

- You have $20,000 in the bank earning interest of 5% per year. You need $30,000 to make a down payment on a house. You can save an additional $1,000 per year, deposited at the end of each year. How long will it take you to accumulate the $30,000?

Practice Problem

- You have \$20,000 in the bank earning interest of 5% per year. You need \$30,000 to make a down payment on a house. You can save an additional \$1,000 per year, deposited at the end of each year. How long will it take you to accumulate the \$30,000?



$I = 5\%, \quad N = ?$

Practice Problem

- Solution, we need to solve for **N**,
  - $-20{,}000$ **PV** (the money already in the bank)
  - $-1{,}000$ **PMT** (the yearly deposits)
  - $30{,}000$ **FV** (the target, with the opposite sign)
  - 5 **I/Y**
  - **CPT**, then **N**

Practice Problem

- Solution, we need to solve for **N**,
  - $-20{,}000$ **PV** (the money already in the bank)
  - $-1{,}000$ **PMT** (the yearly deposits)
  - 30,000 **FV** (the target, with the opposite sign)
  - 5 **I/Y**
  - **CPT**, then **N**
- The calculator is solving

$$-\$20{,}000 - \left[\frac{\$1{,}000}{(1+0.05)^1} + \cdots + \frac{\$1{,}000}{(1+0.05)^N}\right] + \frac{\$30{,}000}{(1+0.05)^N} = 0$$

Practice Problem

- Solution, we need to solve for **N**,
  - $-20{,}000$ **PV** (the money already in the bank)
  - $-1{,}000$ **PMT** (the yearly deposits)
  - 30,000 **FV** (the target, with the opposite sign)
  - 5 **I/Y**
  - **CPT**, then **N**
- The calculator is solving

$$-\$20{,}000 - \left[\frac{\$1{,}000}{(1+0.05)^1} + \cdots + \frac{\$1{,}000}{(1+0.05)^N}\right] + \frac{\$30{,}000}{(1+0.05)^N} = 0$$

- The output is 4.57, so it takes about 4.6 years to accumulate the \$30,000

Module 5, Bonds

- **Bond**, the issuer (the borrower) promises the bondholder (the lender) coupons, then the face value at maturity

Module 5, Bonds

- **Bond**, the issuer (the borrower) promises the bondholder (the lender) coupons, then the face value at maturity
- **Coupon rate** = Annual coupon / Face value. The coupon rate is not the interest rate, a zero-coupon bond still has a yield

Module 5, Bonds

- **Bond**, the issuer (the borrower) promises the bondholder (the lender) coupons, then the face value at maturity
- **Coupon rate** = Annual coupon / Face value. The coupon rate is not the interest rate, a zero-coupon bond still has a yield
- **Bond value**, the present value of the coupons and the face value, discounted at the required rate of return r

$$\text{Value} = \frac{\text{Coupon}}{(1+r)^1} + \frac{\text{Coupon}}{(1+r)^2} + \cdots + \frac{\text{Coupon} + \text{Face value}}{(1+r)^T}$$

Module 5, Bonds

- **Bond**, the issuer (the borrower) promises the bondholder (the lender) coupons, then the face value at maturity
- **Coupon rate** = Annual coupon / Face value. The coupon rate is not the interest rate, a zero-coupon bond still has a yield
- **Bond value**, the present value of the coupons and the face value, discounted at the required rate of return r

$$\text{Value} = \frac{\text{Coupon}}{(1+r)^1} + \frac{\text{Coupon}}{(1+r)^2} + \cdots + \frac{\text{Coupon} + \text{Face value}}{(1+r)^T}$$

- On the calculator, T **N**, r **I/Y**, coupon **PMT**, face value **FV**, then **CPT** **PV**

Required Rate of Return vs. Yield

Required rate of return r

- The discount rate for computing the bond's value
- Chosen by investors as the opportunity cost, the return on alternatives with a similar risk
- Input r, output the value (**CPT** **PV**)

Yield y

Required Rate of Return vs. Yield

Required rate of return r

- The discount rate for computing the bond's value
- Chosen by investors as the opportunity cost, the return on alternatives with a similar risk
- Input r, output the value (**CPT** **PV**)

Yield y

- The discount rate for which the present value of the payments equals the price
- Implied by the market price
- Input the price, output the yield (**CPT** **I/Y**)

Required Rate of Return vs. Yield

Required rate of return r

- The discount rate for computing the bond's value
- Chosen by investors as the opportunity cost, the return on alternatives with a similar risk
- Input r, output the value (CPT PV)
- **Subjective**

Yield y

- The discount rate for which the present value of the payments equals the price
- Implied by the market price
- Input the price, output the yield (CPT I/Y)
- **Objective**

Module 5, Bond Prices and Yields

- Bond prices and yields move in opposite directions

Module 5, Bond Prices and Yields

- Bond prices and yields move in opposite directions
- What determines the price (and so the yield)? Market interest rates and the risk of default

Module 5, Bond Prices and Yields

- Bond prices and yields move in opposite directions
- What determines the price (and so the yield)? Market interest rates and the risk of default
- Bond yield = risk-free rate + default premium, Treasury bonds have no default risk

Module 5, Bond Prices and Yields

- Bond prices and yields move in opposite directions
- What determines the price (and so the yield)? Market interest rates and the risk of default
- Bond yield = risk-free rate + default premium, Treasury bonds have no default risk
- Credit ratings capture default risk, investment-grade (Baa or BBB and higher) against high-yield ("junk") bonds, lower-rated bonds carry higher yields and lower prices

Module 5, Stock Valuation, Dividend Discount Model

- Stock value, the present value of the expected dividends, discounted at the required return on equity r_e

$$P = \frac{\text{Div}_1}{(1+r_e)^1} + \frac{\text{Div}_2}{(1+r_e)^2} + \frac{\text{Div}_3}{(1+r_e)^3} + \cdots$$

Module 5, Stock Valuation, Dividend Discount Model

- Stock value, the present value of the expected dividends, discounted at the required return on equity r_e

$$P = \frac{\text{Div}_1}{(1+r_e)^1} + \frac{\text{Div}_2}{(1+r_e)^2} + \frac{\text{Div}_3}{(1+r_e)^3} + \cdots$$

- Constant dividends, $P = \text{Div}_1/r_e$. Constant growth (the Gordon growth model), $P = \text{Div}_1/(r_e - g)$

Module 5, Stock Valuation, Dividend Discount Model

- Stock value, the present value of the expected dividends, discounted at the required return on equity r_e

$$P = \frac{\text{Div}_1}{(1+r_e)^1} + \frac{\text{Div}_2}{(1+r_e)^2} + \frac{\text{Div}_3}{(1+r_e)^3} + \cdots$$

- Constant dividends, $P = \text{Div}_1/r_e$. Constant growth (the Gordon growth model), $P = \text{Div}_1/(r_e - g)$
- Non-constant growth, discount the dividends year by year up to a horizon H, then the price $P_H = \text{Div}_H \times (1+g)/(r_e - g)$

Module 5, Stock Valuation, Dividend Discount Model

- Stock value, the present value of the expected dividends, discounted at the required return on equity r_e

$$P = \frac{\text{Div}_1}{(1+r_e)^1} + \frac{\text{Div}_2}{(1+r_e)^2} + \frac{\text{Div}_3}{(1+r_e)^3} + \cdots$$

- Constant dividends, $P = \text{Div}_1/r_e$. Constant growth (the Gordon growth model), $P = \text{Div}_1/(r_e - g)$
- Non-constant growth, discount the dividends year by year up to a horizon H, then the price $P_H = \text{Div}_H \times (1+g)/(r_e - g)$
- **Earnings per share (EPS)** = Net income / Number of shares outstanding. **P/E ratio** = Stock price / EPS

Example, Valuing Non-Constant Growth

Recall from **Modules 5 and 6**

- A stock's expected dividends are in the table. Its required return on equity is 10%. Its dividends grow at 5% a year beginning in year 6. What is the value of the stock?

Year	1	2	3	4	5
Dividend	$2	$2	$2	$3	$3

Example, Valuing Non-Constant Growth

Recall from **Modules 5 and 6**

- A stock's expected dividends are in the table. Its required return on equity is 10%. Its dividends grow at 5% a year beginning in year 6. What is the value of the stock?

Year	1	2	3	4	5
Dividend	\$2	\$2	\$2	\$3	\$3

- We discounted the year-5 price of the constant-growth stream, then the five dividends, term by term,

$$\text{PV}_{\text{Price in year 5}} = \frac{3 \times 1.05}{0.10 - 0.05} \times \frac{1}{1.10^5} = \$39.12$$

Example, Valuing Non-Constant Growth

Recall from **Modules 5 and 6**

- A stock's expected dividends are in the table. Its required return on equity is 10%. Its dividends grow at 5% a year beginning in year 6. What is the value of the stock?

Year	1	2	3	4	5
Dividend	\$2	\$2	\$2	\$3	\$3

- We discounted the year-5 price of the constant-growth stream, then the five dividends, term by term,

$$\text{PV}_{\text{Price in year 5}} = \frac{3 \times 1.05}{0.10 - 0.05} \times \frac{1}{1.10^5} = \$39.12$$

$$\text{PV}_{\text{Dividends 1–5}} = \frac{2}{1.10} + \frac{2}{1.10^2} + \frac{2}{1.10^3} + \frac{3}{1.10^4} + \frac{3}{1.10^5} = \$8.89$$

Example, Valuing Non-Constant Growth

Recall from **Modules 5 and 6**

- A stock's expected dividends are in the table. Its required return on equity is 10%. Its dividends grow at 5% a year beginning in year 6. What is the value of the stock?

Year	1	2	3	4	5
Dividend	\$2	\$2	\$2	\$3	\$3

- We discounted the year-5 price of the constant-growth stream, then the five dividends, term by term,

$$\text{PV}_{\text{Price in year 5}} = \frac{3 \times 1.05}{0.10 - 0.05} \times \frac{1}{1.10^5} = \$39.12$$

$$\text{PV}_{\text{Dividends 1–5}} = \frac{2}{1.10} + \frac{2}{1.10^2} + \frac{2}{1.10^3} + \frac{3}{1.10^4} + \frac{3}{1.10^5} = \$8.89$$

- Total, $\text{PV}_{\text{Stock}} = 39.12 + 8.89 = \$48.00$

Example, Non-Constant Growth with a Financial Calculator



$$+ \frac{3 \times 1.05}{0.10 - 0.05}$$

Example, Non-Constant Growth with a Financial Calculator



$$+\frac{3\times 1.05}{0.10-0.05}$$

- Enter the cash flows one by one, 0, 2, 2, 2, 3, 66
- In year 5, enter the dividend plus the price, $3+63=66$

Enter	Keys
	CF, **2nd**, **CLR Work**
0	**ENTER**, ↓
2	**ENTER**, ↓ ↓
2	**ENTER**, ↓ ↓
2	**ENTER**, ↓ ↓
3	**ENTER**, ↓ ↓
66	**ENTER**
	NPV (the calculator asks for I)
10	**ENTER**, ↓
	CPT
48.00	

Example, Non-Constant Growth with a Financial Calculator



$$+\frac{3 \times 1.05}{0.10 - 0.05}$$

- Enter the cash flows one by one, 0, 2, 2, 2, 3, 66
- In year 5, enter the dividend plus the price, $3 + 63 = 66$
- **No need to discount each cash flow and sum by hand!**

Enter	Keys
	CF, **2nd**, **CLR Work**
0	**ENTER**, ↓
2	**ENTER**, ↓ ↓
2	**ENTER**, ↓ ↓
2	**ENTER**, ↓ ↓
3	**ENTER**, ↓ ↓
66	**ENTER**
	NPV (the calculator asks for I)
10	**ENTER**, ↓
	CPT
48.00	

Example, Buying the Stock Is an NPV Decision

- The same stock (dividends 2, 2, 2, 3, 3, then 5% growth, required return on equity 10%) sells for $50 today
- Should you buy it?

Example, Buying the Stock Is an NPV Decision

- The same stock (dividends 2, 2, 2, 3, 3, then 5% growth, required return on equity 10%) sells for $50 today
- Should you buy it?
- Solution, buying the stock is a project, the initial investment is $50 and the future cash flows are the dividends

Example, Buying the Stock Is an NPV Decision

- The same stock (dividends 2, 2, 2, 3, 3, then 5% growth, required return on equity 10%) sells for \$50 today
- Should you buy it?
- Solution, buying the stock is a project, the initial investment is \$50 and the future cash flows are the dividends

$$\text{NPV} = -50 + \text{PV(Dividends)} = -50 + 48.00 = -\$2.00$$

Example, Buying the Stock Is an NPV Decision

- The same stock (dividends 2, 2, 2, 3, 3, then 5% growth, required return on equity 10%) sells for \$50 today
- Should you buy it?
- Solution, buying the stock is a project, the initial investment is \$50 and the future cash flows are the dividends

$$\text{NPV} = -50 + \text{PV}(\text{Dividends}) = -50 + 48.00 = -\$2.00$$

- You should not buy the stock, because the NPV is negative

Example, Buying the Stock Is an NPV Decision

- The same stock (dividends 2, 2, 2, 3, 3, then 5% growth, required return on equity 10%) sells for \$50 today
- Should you buy it?
- Solution, buying the stock is a project, the initial investment is \$50 and the future cash flows are the dividends

$$\text{NPV} = -50 + \text{PV(Dividends)} = -50 + 48.00 = -\$2.00$$

- You should not buy the stock, because the NPV is negative
- On the calculator, the same worksheet with -50 entered as the first cash flow (instead of 0)

Module 5, Stock Valuation, Comparables

- Comparables, value a stock by "scaling" the prices of similar companies

Module 5, Stock Valuation, Comparables

- Comparables, value a stock by "scaling" the prices of similar companies
- Average the comparables' multiple (P/E or market-to-book), then multiply by the company's own denominator,

$$P_X = (\text{Average P/E of comparables}) \times \text{EPS}_X$$

Module 5, Stock Valuation, Comparables

- Comparables, value a stock by "scaling" the prices of similar companies
- Average the comparables' multiple (P/E or market-to-book), then multiply by the company's own denominator,

$$P_X = (\text{Average P/E of comparables}) \times \text{EPS}_X$$

- With market-to-book, Market capitalization = (Average M/B of comparables) $\times$ Book value of equity, then P = Market capitalization / Number of shares outstanding

Module 5, Stock Valuation, Comparables

- Comparables, value a stock by "scaling" the prices of similar companies
- Average the comparables' multiple (P/E or market-to-book), then multiply by the company's own denominator,

$$P_X = (\text{Average P/E of comparables}) \times \text{EPS}_X$$

- With market-to-book, Market capitalization = (Average M/B of comparables) $\times$ Book value of equity, then P = Market capitalization / Number of shares outstanding
- The estimate is sensitive to the choice of multiple, and good comparables are hard to find

Capital Budgeting Techniques, Summary

Criterion	Definition	Investment rule
Net present value (NPV)	Present value of benefits minus present value of costs	Take all projects with NPV > 0, reject all with NPV < 0
Internal rate of return (IRR)	The discount rate at which NPV $= 0$	Accept if IRR $>$ opportunity cost r, reject if IRR $< r$
Profitability index	Net present value divided by initial investment	Under capital rationing, rank projects by their profitability index and go down the ranking
Payback period	The number of years until the cash flows recover the initial investment	Accept if the payback period is shorter than a maximum cutoff

Facing Multiple Projects, Summary

Situation	What to do
No constraint	Take all projects with NPV $>$ 0, reject all with NPV $<$ 0
Can only choose one among mutually exclusive projects	Take the positive-NPV project with the highest NPV, not the highest IRR
Limited amount of funds for the initial investment (capital rationing)	Rank projects by their profitability index from highest to lowest and go down the ranking, stop when you run out of dollars or run out of positive-NPV projects

Thank you!

Please let me know if you have any questions.